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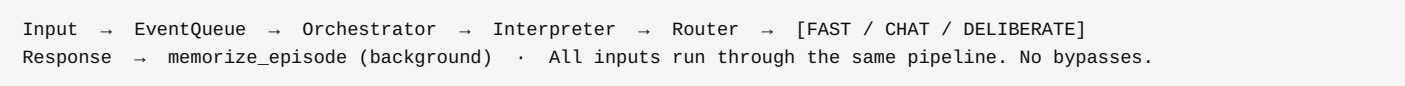
C.L.A.R.A.

Contextual Locally Aware Robust Agent
Python · FastAPI · React · DeepSeek API · FAISS · SQLite · RTX 3050 (4GB VRAM)

WHAT IT IS

CLARA is a personal autonomous AI system built on a simple premise: a capable AI shouldn't need to be invoked for every task. Every component is purpose-built rather than assembled from existing frameworks. The goal was a system that stays present — maintaining context across sessions, running background processes on its own, and routing tasks by reasoning complexity rather than keyword matching. It doesn't just respond. It operates.

SYSTEM FLOW



CAPABILITIES

Execution pipeline	FAST/CHAT/DELIBERATE routing by confidence and complexity. Parallel tool batching via <code>asyncio.gather()</code> . FAST auto-escalates to DELIBERATE on failure with full failure context injected.
Memory system	Semantic episodic retrieval combining recency and cosine similarity. Vault with 0.85 cosine dedup. Style persistence across sessions.
Knowledge base	FAISS RAG indexed on live documentation. Auto-rebuild on source file changes. Hot-reload without restart. Passive archive injection per request.
Autonomy	Continuous background loop with scheduled tasks, environment triggers, interrupt handling, and crash recovery via SQLite.
Tool ecosystem	Native tools (web search, Python REPL, vision, RAG) plus MCP via Desktop Commander — 24 tools. Semantic tool registry finds the right tool, not just the one you named.
Voice & Interface	Push-to-talk via Faster-Whisper STT + Kokoro TTS (~200ms latency). Live task board, streaming thought panel, animated vitals over WebSocket.

WHAT SETS IT APART

Most agent systems today are assembled from existing primitives — they work, but the architecture shows. CLARA's memory, knowledge base, autonomy, and reasoning share a single event-driven pipeline because they need to. Context that doesn't flow between components isn't really context. The design goal wasn't capability breadth but coherence: a system that knows what it's doing, why, and what it did last time.

WHERE IT APPLIES

On-device personal AI	Always-on agent with persistent memory. No cloud dependency.
Enterprise knowledge worker	Autonomous document processing and task orchestration over internal systems.
Developer assistant	Codebase-aware agent with file system access, tool execution, and project memory.

SCOPE

35 implementation briefs across ~30 source files. Full pipeline in daily use — voice, memory, tool registry, and resource safety all live.

WHAT'S NEXT

- Streaming migration — single LLM call with mid-stream tool execution; eliminates multi-turn overhead
- Ambient awareness — JARVIS-style unprompted observations based on live context
- External events — WhatsApp/calendar as EventQueue inputs
- Life context layer — structured fields for fitness, goals, habits

